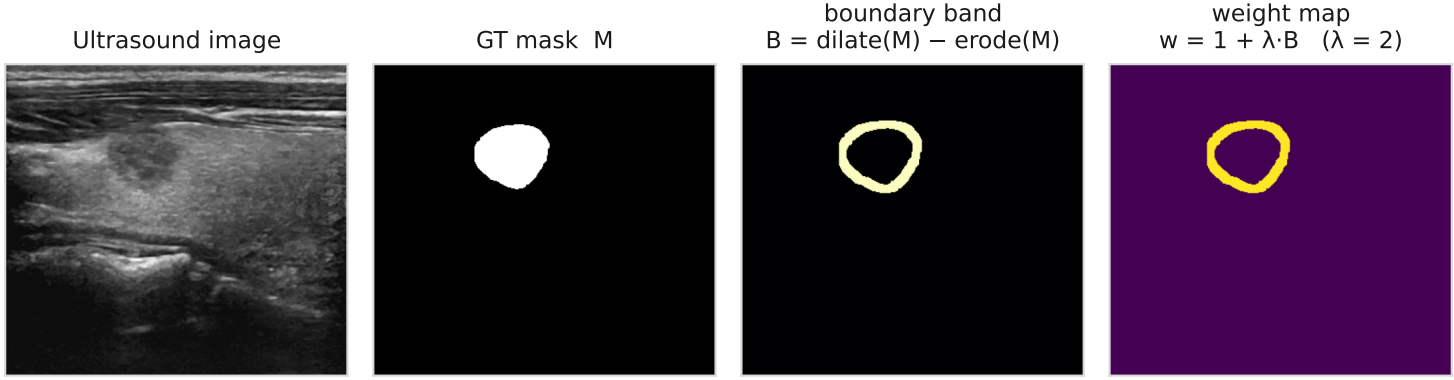
Boundary-aware loss: BCE on mask pixels is up-weighted near the ground-truth boundary



$$\mathcal{L}_{mask} = \frac{1}{N}\sum_{p} w_p \cdot \mathrm{BCE}(\hat{m}_p, m_p) \quad \cdot \quad \mathcal{L}_{edge} = \mathrm{BCE}(\hat{e}, B) \quad \cdot \quad \mathcal{L}_{seg} = \mathcal{L}_{mask} + \gamma\,\mathcal{L}_{edge} \quad (\gamma = 1.0)$$